

National University of Computer and Emerging Sciences, Lahore Campus



Course:
Program:
Date:
Section:
Assignment no. :

Data Mining
BSE
13-02-26
6A
01

Total Marks:
Name:
Roll #:

Important Instructions:

- Use A4 sheets or narrow line sheets for solving assignments.
- Assignments done on register pages will be cancelled.
- Must attach the assignment title page shared to you at the end of assignment.
- This assignment must be completed independently by the student. The use of AI tools (including ChatGPT, Bard, Copilot, or any other generative AI system) is strictly prohibited.
- Students are expected to show all manual calculations and intermediate steps clearly.

Q1 Outliers are often discarded as noise. However, one person's garbage could be another's treasure. For example, exceptions in credit card transactions can help us detect the fraudulent use of credit cards. Using fraudulence detection as an example, propose two methods that can be used to detect outliers and discuss which one is more reliable.

Q2 Suppose that the data for analysis includes the attribute age. The age values for the data tuples are (in increasing order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.

(a) What is the mean of the data? What is the median?

(b) What is the mode of the data? Comment on the data's modality (i.e., bimodal, trimodal, etc.).

(c) What is the midrange of the data?

(d) Can you find (roughly) the first quartile (Q1) and the third quartile (Q3) of the data?

(e) Give the five-number summary of the data.

(f) Show a boxplot of the data.

(g) How is a quantile-quantile plot different from a quantile plot?

Q3 Suppose a hospital tested the age and body fat data for 18 randomly selected adults with the following result

age	23	23	27	27	39	41	47	49	50
%fat	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2
age	52	54	54	56	57	58	58	60	61
%fat	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

- (a) Calculate the mean, median and standard deviation of age and %fat.
- (b) Draw the boxplots for age and %fat.
- (c) Draw a scatter plot and a q-q plot based on these two variables.

Q4 6. Given two objects represented by the tuples (22, 1, 42, 10) and (20, 0, 36, 8):

(a) Compute the Euclidean distance between the two objects.

(b) Compute the Manhattan distance between the two objects.

(c) Compute the Minkowski distance between the two objects, using $h = 3$.

(d) Compute the supremum distance between the two objects.

Q5

Given is a term frequency vector. Calculate the cosine similarity between
 i. document 1 and document 2
 ii. document 3 and document 4

Doc	Team	Coach	hockey	Ball	soccer	Penalty	score	Win	loss	season
Doc 1	5	0	3	0	2	0	0	2	0	0
Doc 2	3	0	2	0	1	1	0	1	0	1
Doc 3	0	7	0	2	1	0	0	3	0	0
Doc 4	0	1	0	0	1	2	2	0	3	0

Q6

Classify the following attributes as binary, discrete, or continuous. Also classify them as qualitative (nominal or ordinal) or quantitative (interval or ratio). Some cases may have more than one interpretation, so briefly indicate your reasoning if you think there may be some ambiguity.

Example: Age in years. Answer: Discrete, quantitative, ratio

(a) Time in terms of AM or PM.

(b) Brightness as measured by a light meter.

(c) Brightness as measured by people's judgments.

(d) Angles as measured in degrees between 00 and 3600

(e) Bronze, Silver, and Gold medals as awarded at the Olympics.

(f) Height above sea level.

(g) Number of patients in a hospital.

(h) ISBN numbers for books. (Look up the format on the Web.)

(i) Ability to pass light in terms of the following values: opaque, translucent, transparent.

(j) Military rank.

(k) Distance from the center of campus.

(l) Density of a substance in grams per cubic centimeter.

(m) Coat check number. (When you attend an event, you can often give your coat to someone who, in turn, gives you a number that you can use to claim your coat when you leave.)

Q7

Compute the distance matrix using Euclidean and Manhattan formula

points	p1	p2	p3	p4	p5	p6
x	0.40	0.22	0.35	0.26	0.08	0.45
y	0.53	0.38	0.32	0.19	0.41	0.30

Q8

Calculate Jacquard Coefficient and Extended Jacquard Coefficient for the given data.

P	1	0	0	0	0	1	1	1	0	0
Q	0	0	0	1	1	0	0	1	1	0

Q9 Run the apriori algorithm on the given dataset.

TID	Items Purchased
T1	Milk, Bread, Butter
T2	Bread, Butter
T3	Milk, Bread
T4	Milk, Butter
T5	Bread, Diapers
T6	Milk, Bread, Diapers
T7	Milk, Bread, Butter, Diapers
T8	Bread, Butter
T9	Milk, Diapers
T10	Milk, Bread, Butter

Q10 Compute PCA for given data. You are required to reduce the dimensionality from 4 dimensions to 2 dimensions.

Features	E1	E2	E3	E4	E5	E6	E7	E8	E9	E10
X1	2	3	4	5	6	7	8	9	1	5
X2	3	6	4	6	8	9	9	2	5	6
X3	5	6	7	8	2	3	4	5	4	3
X4	6	8	9	9	3	6	4	6	7	7

Department of Computer Science, School of Computing

DATA MINING



Assignment # _____

Section: _____

Submitted by:

Name: _____

Roll no: _____

Submitted to:

Dr. Muhammad Aasim Qureshi

Date Assigned: _____

Date Submission: _____